

LYRA: GENERATIVE 3D SCENE RECONSTRUCTION VIA VIDEO DIFFUSION MODEL SELF-DISTILLATION

Anonymous authors
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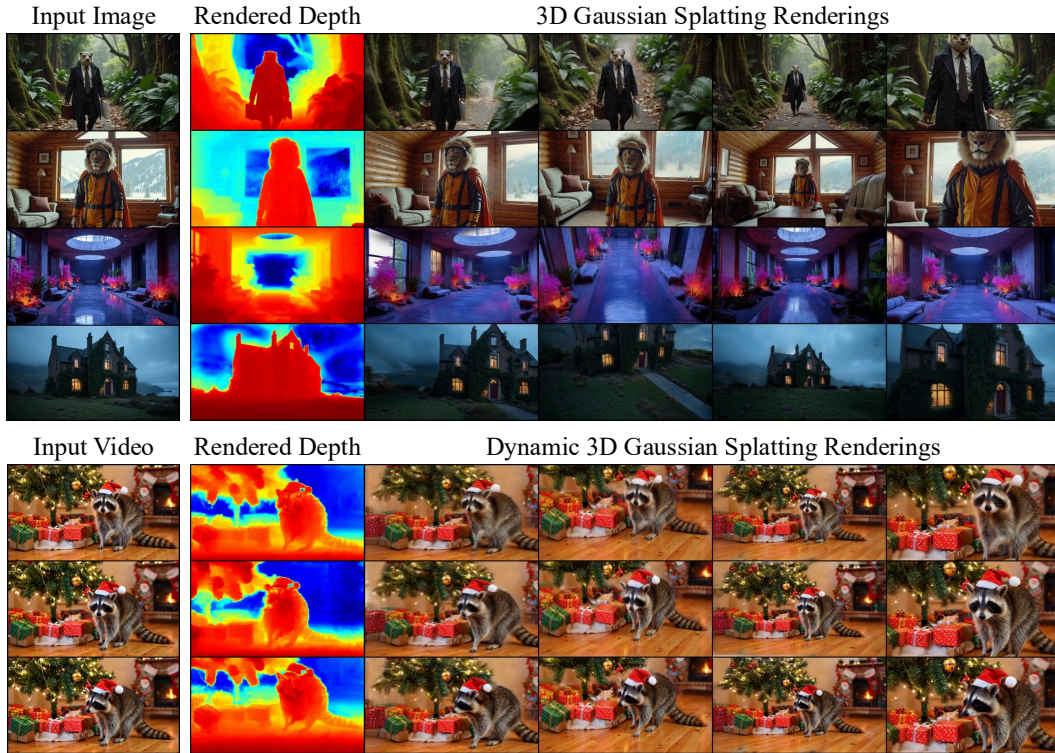


Figure 1: **Feed-Forward 3D and 4D Scene Generation.** From a single image (top), *Lyra* infers a 3D Gaussian Splatting (3DGS) representation in a single feed-forward pass by self-distilling a video diffusion model, with no real-world multi-view data. From a video input (bottom), *Lyra* infers a dynamic 3DGS that delivers interactive control over both time (rows) and viewpoint (columns).

ABSTRACT

Generating virtual environments is a cornerstone capability for applications ranging from gaming to physical AI domains such as robotics, autonomous driving, and industrial AI. Current learning-based 3D reconstruction methods are fundamentally bottlenecked by their dependence on captured real-world multi-view data, which is rarely available at scale. Video diffusion models exhibit remarkable imagination capabilities, yet their purely 2D nature has so far ruled them out of simulation settings where a robot must navigate and interact with the environment. In this paper, we resolve this gap with a self-distillation framework that distills the implicit 3D knowledge embedded in video diffusion models into an explicit 3D Gaussian Splatting (3DGS) representation, entirely removing the need for multi-view training data. Concretely, we augment the standard RGB decoder with a 3DGS decoder that is directly supervised by the RGB decoder’s output. As a result, the 3DGS decoder is trained purely on synthetic data generated by video diffusion models—no real captures required. At inference time, our model synthesizes 3D scenes from either a text prompt or a single image and renders them in real time. The framework extends seamlessly to dynamic 3D scene generation from a monocular input video.

Experiments establish that our framework sets a new state of the art in both static and dynamic 3D scene generation.

1 INTRODUCTION

Creating high-quality 3D environments at scale is a long-standing challenge in computer vision and graphics, and solving it directly empowers applications across film-making, VR/AR, and physical AI with closed-loop simulation. These applications demand explicit 3D representations that support real-time rendering, physical interaction, and consistent multi-view synthesis.

Recent advances in neural 3D reconstruction (Mildenhall et al., 2020; Kerbl et al., 2023; Moenne-Loccoz et al., 2024) recover such representations from posed images, but their reliance on accurate camera poses and high-quality images is a decisive barrier to scalability. The challenge is even sharper for dynamic scenes, which typically require synchronized multi-camera setups (Li et al., 2022). Reconstruction methods are also inherently confined to observed content and cannot extrapolate beyond the input views. Feed-forward reconstruction models (Zhang et al., 2024e; Ren et al., 2024c) were introduced to relax these limitations, but they simply relocate the bottleneck: the scarcity of diverse, large-scale 3D training data leaves them with poor out-of-domain generalization.

Video diffusion models (Cosmos, 2025; Wan, 2025) offer a decisively better foundation. Trained on massive internet-scale video corpora, they achieve strong fidelity and generalization across diverse environments. By learning from real-world videos with varied camera trajectories—handheld motion, cinematic panning, and drone footage—these models implicitly encode rich cues about the underlying 3D world without any multi-view training data. Crucially, unlike reconstruction approaches (Charatan et al., 2024), generative models also hallucinate plausible content well beyond what is visible in the input frames. Their one limitation is that they emit only 2D frames and no explicit 3D representation, which has kept them out of simulation and downstream tasks that demand geometric consistency, long-term coherence, and physical interaction. Recent work (Wang et al., 2024e; Ren et al., 2025) removes even this obstacle, showing that video models can be equipped with explicit camera control using only small-scale posed datasets. The ability to generate posed image sequences turns these models into a powerful engine that serves as both input and supervision for 3D reconstruction.

Building on this insight, we unify these two paradigms, reconstruction and generation, into a single framework we call *generative 3D scene reconstruction*. We introduce *Lyra*, a method that generates explicit 3D environments directly from the latent representations of video diffusion models in a single forward pass. At its core, *Lyra* is trained in a self-distillation framework (Fig. 2), with a 3D Gaussian Splatting (3DGS) decoder that operates directly in the latent space of a video diffusion model. Given a single input image or video, we sample a camera trajectory as conditioning, denoise the resulting video latent, and decode it along two parallel branches. The first branch uses the standard RGB decoder to synthesize a video sequence; the second is our 3DGS decoder, which produces an explicit 3D representation. Together these branches form a self-distillation loop in which the RGB branch (teacher) supervises the 3DGS branch (student). To broaden viewpoint coverage, we sample multiple camera trajectories, generate multiple video latents, and train the 3DGS decoder to fuse information across them while resolving long-term and multi-view inconsistencies.

This generative reconstruction framework delivers three decisive benefits: **(i)** it generates large-scale synthetic environments spanning diverse scenarios directly from video diffusion models, eliminating any need for real-world multi-view captures; **(ii)** by operating in the latent space of a video model, it processes many views efficiently, sidestepping the heavy memory overhead that cripples pixel-space feed-forward reconstruction methods; and **(iii)** its explicit 3DGS output guarantees geometric consistency, yielding representations that are immediately usable in downstream tasks such as physical simulation. As a direct consequence, at inference time *Lyra* generates high-quality 3D Gaussian scenes from monocular input that render in real time, *without any additional optimization or post-processing*.

We further extend this self-distillation framework to dynamic 4D generation from monocular video. Here the video model (teacher) supplies space–time supervision, while the student learns time-conditioned 3DGS representations that enable novel-view synthesis of dynamic scenes.

In summary, this work makes the following contributions:

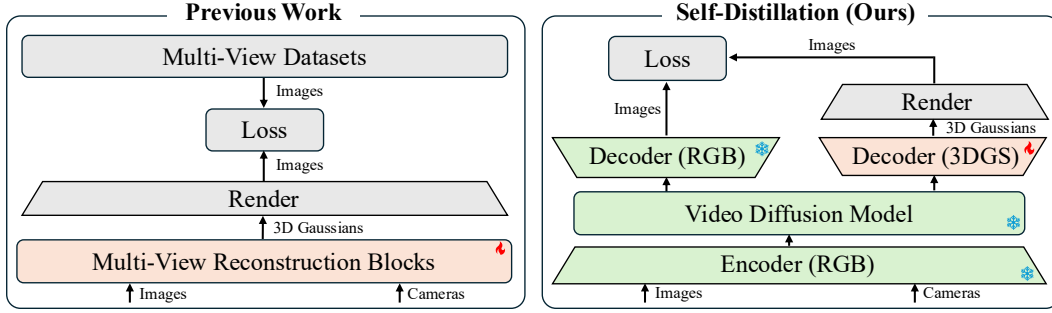


Figure 2: **Self-distillation framework of *Lyra***. Previous work (left) (Szymanowicz et al., 2025b) trains multi-view reconstruction blocks on real-world datasets of limited diversity (Zhou et al., 2018; Ling et al., 2024b). We instead introduce a self-distillation framework (right) for generative 3D scene reconstruction: a pre-trained camera-controlled video diffusion model, through its RGB decoder output (teacher), supervises the rendering of the 3DGS decoder (student). Because the video model is pre-trained on diverse 2D video data, it supplies far more diverse multi-view supervision than real captures can.

- We introduce a self-distillation framework that trains a 3DGS student decoder from a pre-trained camera-controlled video diffusion model as teacher, eliminating the need for captured multi-view real-world data.
- We show that this framework extends naturally to dynamic scenes, enabling 4D reconstruction from a single monocular video.
- We demonstrate that our model generalizes across diverse scenes, setting a new state of the art in both single-image 3D scene generation and single-video 4D scene generation.

2 RELATED WORK

Multi-view image generation. Early works on multi-view image generation (Watson et al., 2023; Liu et al., 2023; Shi et al., 2024; Wang & Shi, 2023) mainly focus on object-centric scenes without background. Follow-up works (Sargent et al., 2024; Tang et al., 2023b; Schneider et al., 2025) extend multi-view image generators to the scene scale. CAT3D (Gao et al., 2024b) extends a pre-trained image diffusion model for multi-view image generation from any generated or real image input. In a subsequent stage, the generated multi-view images are reconstructed into an explicit 3D representation, e.g., NeRF (Mildenhall et al., 2020) or 3DGS (Kerbl et al., 2023). Moreover, these methods have been extended to generate dynamic 3D scenes from multiple viewpoints using a space-time diffusion model (Watson et al., 2024; Wu et al., 2025b; Kuang et al., 2024; Wang et al., 2025a). Impressive as these results are, grounding the generations in an explicit 3D representation forces an expensive optimization stage that is not amortized across scenes. Rather than creating multi-view images and then reconstructing them through optimization, we *directly* generate a 3D scene from text or a single image.

Camera-conditioned video generation. There has been significant progress in fine-tuning video diffusion models for 3D camera control. MotionCtrl (Wang et al., 2024e) pioneered camera control by conditioning pre-trained video models with camera poses. Follow-up works (He et al., 2024a; Xu et al., 2024c; Bahmani et al., 2025a;b) represent cameras as Plücker coordinates for pixel-wise conditioning. Another line of work (Hu et al., 2024; Hou et al., 2024; Zhang et al., 2024b) investigates training-free camera control of video diffusion models. Recently, SynCamMaster (Bai et al., 2025b) and ReCamMaster (Bai et al., 2025a) demonstrated impressive camera-controlled video synthesis using large-scale synthetic training data. Compelling as these works are, their output is not grounded in 3D. Our work is orthogonal: we tackle feed-forward 3D generation from camera-controlled video models, enabling novel-viewpoint rendering *after* generation. Concretely, we build upon GEN3C (Ren et al., 2025), a recent camera-controlled video diffusion model, and repurpose it as a teacher within a self-distillation framework for 3D reconstruction.

Feed-forward 3D models. Early work on feed-forward 3D reconstruction mainly focused on object-centric scenes without background. These methods directly predict a NeRF (Hong et al., 2024; Li et al., 2024b) or 3DGS (Tang et al., 2024a; Szymanowicz et al., 2024) from either text or an image. More recent works (Charatan et al., 2024; Zhang et al., 2024e; Szymanowicz et al., 2025a; Ren et al., 2024c) focus on scene-level 3D reconstruction, typically regressing per-pixel 3D

Gaussians from one or more input images. However, most scene-level 3D reconstruction methods remain confined to their training distribution, e.g., RealEstate10K (Zhou et al., 2018), and generalize poorly to generated scenes. Recent methods push feed-forward 3D reconstruction toward scene generation. Bolt3D (Szymanowicz et al., 2025b) trains a pointmap (Wang et al., 2024b) autoencoder to generate pointmaps alongside the multi-view images for feed-forward 3D reconstruction. Most closely related to us, Wonderland (Liang et al., 2025a) uses a camera-controlled video model to generate 3D Gaussians with a feed-forward network. In sharp contrast to these works, we drive a camera-controlled video model as a teacher within a self-distillation framework to train our student 3D model *without any real-world multi-view data*. Crucially, we further show a simple extension of our framework to feed-forward 4D scene generation—a task left unexplored to date.

3 SELF-DISTILLATION USING VIDEO DIFFUSION MODELS

Our central idea is to distill the implicit 3D knowledge embedded in video diffusion models into an explicit 3DGS decoder that produces high-quality 3D representations. We realize this with a teacher–student framework in which the video diffusion model (teacher) generates RGB videos that supervise the 3DGS decoder (student) operating in the same latent space, as illustrated in Fig. 2. In this section, we detail our video diffusion model backbone (Sec. 3.1) and its role in self-distillation for the 3DGS decoder (Sec. 3.2). We then present the design of the 3DGS decoder in Sec. 4 and show that only minimal changes are needed to adapt the pipeline to dynamic 3D scenes in Sec. 5.

3.1 BACKGROUND: CAMERA-CONTROLLED AND 3D-CONSISTENT VIDEO DIFFUSION

Because our 3DGS decoder is supervised solely by the camera-controlled video diffusion model, its 3D consistency is paramount. We therefore build our approach on GEN3C (Ren et al., 2025), a recent camera-conditioned video diffusion model that delivers exactly this property. Below we review the key design choices behind GEN3C.

Spatiotemporal 3D cache. To improve video consistency and camera control precision, GEN3C constructs a spatiotemporal 3D cache $\{\mathbf{P}^{t,v}\}$ from input image(s) or videos, where each $\mathbf{P}^{t,v}$ is a colored point cloud obtained by unprojecting the depth estimation (Wang et al., 2024a) of an RGB image captured from camera viewpoint v at time t . The cache is organized as an $L \times V$ array, where L denotes the number of frames (temporal length) and V denotes the number of camera views.

Rendering and structured guidance. To leverage this cache, GEN3C renders each point cloud from arbitrary given camera poses, producing RGB images $\mathbf{I}^{t,v}$ and disocclusion masks $\mathbf{M}^{t,v}$ via $(\mathbf{I}^{t,v}, \mathbf{M}^{t,v}) = \mathcal{R}(\mathbf{P}^{t,v}, \mathbf{C}^t)$, where $\mathcal{R}$ denotes the rendering function that projects the 3D point cloud $\mathbf{P}^{t,v}$ onto the 2D camera plane according to the camera pose $\mathbf{C}^t$ at time t . Given a sequence of camera poses $\mathbf{C} = (\mathbf{C}^1, \dots, \mathbf{C}^L)$, rendering all cache elements produces V videos of length L , which can be stacked into image sequences $\mathbf{I}^v \in \mathbb{R}^{L \times 3 \times H \times W}$ and mask sequences $\mathbf{M}^v \in \mathbb{R}^{L \times 1 \times H \times W}$ for each view v . The disocclusion masks indicate areas that the video diffusion model should fill in. These renderings serve as structured visual guidance for subsequent video generation.

Video variational autoencoder (VAE). In diffusion-based video generation models, a video variational autoencoder (VAE) (Kingma & Welling, 2013; Rombach et al., 2022) is commonly employed to compress videos into a lower-dimensional latent space for efficient training and inference. Given a RGB video $\mathbf{I} \in \mathbb{R}^{L \times 3 \times H \times W}$, a pre-trained VAE encoder $\mathcal{E}$ will encode the video into a latent space, i.e. $\mathbf{z} = \mathcal{E}(\mathbf{I}) \in \mathbb{R}^{L' \times C \times h \times w}$. The training and inference of the diffusion model are then performed in this latent space. The final video $\hat{\mathbf{I}} = \mathcal{D}_{rgb}(\mathbf{z})$ is decoded with a pre-trained VAE decoder $\mathcal{D}_{rgb}$. In this paper, we adopt the pre-trained GEN3C (Ren et al., 2025; Cosmos, 2025) model. Specifically, the latent channel dimension is $C = 16$, the temporal dimension is $L' = (L - 1)/\tau + 1$, and the spatial dimensions are $h = H/\sigma$ and $w = W/\sigma$, where the temporal compression factor is $\tau = 8$ and the spatial compression factor is $\sigma = 8$.

3.2 SELF-DISTILLATION

We train the 3DGS decoder $\mathcal{D}_s$ as a student under a teacher–student paradigm in which a camera-controlled video diffusion model $\mathcal{V}$ acts as the teacher. To secure diverse supervision, we curate a large-scale set of text prompts with large language models (OpenAI, 2025; Bai et al., 2023), generate

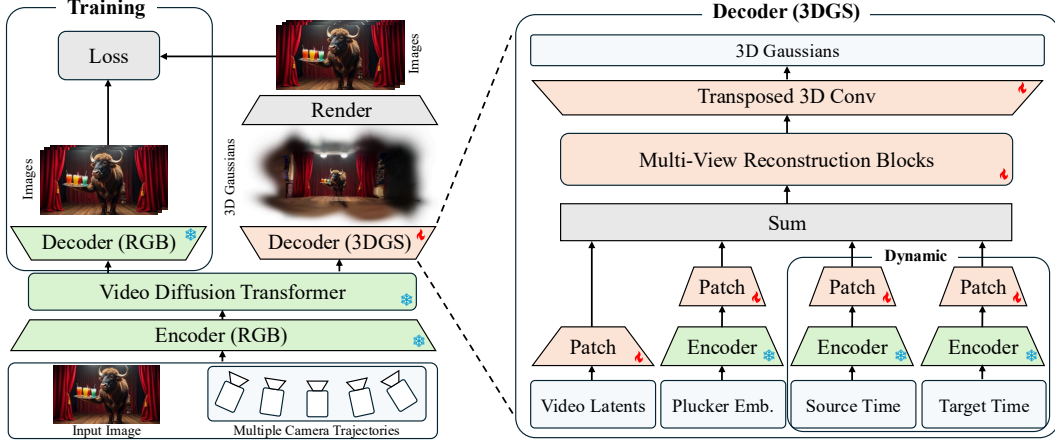


Figure 4: **3D Generative Reconstruction Framework.** Our pipeline builds upon a camera-controlled video diffusion model (Ren et al., 2025) pre-trained on large-scale data. We train a 3D Gaussian Splatting (3DGS) decoder by aligning the 2D image renderings of generated 3DGS scenes with the RGB-decoded generations of the pre-trained video model, training only the 3DGS decoder while keeping the pre-trained autoencoder and diffusion model frozen. At inference time we use the 3DGS decoder alone, dropping the RGB decoder entirely. Time conditioning within the 3DGS decoder lets us extend the approach from static to dynamic 3D scene generation with minimal effort.

images I with an image diffusion model (Black Forest Labs, 2024), and expand each into multi-view sequences using GEN3C (Ren et al., 2025).

Teacher–student setup. Given an input image I and a sampled camera trajectory $\{\mathbf{C}^t\}_{t=1}^L$, the video diffusion model $\mathcal{V}$ generates a denoised video latent $\mathbf{z} = \mathcal{V}(I, \{\mathbf{C}^t\}_{t=1}^L)$. The latent $\mathbf{z}$ is decoded along two branches: the pre-trained RGB decoder $\mathcal{D}_{rgb}$ produces video frames $\mathbf{I}_{\mathcal{D}_{rgb}} = \mathcal{D}_{rgb}(\mathbf{z})$, while the 3DGS decoder $\mathcal{D}_s$ outputs explicit 3D Gaussians $\mathbf{G}$. Rendered views from 3DGS (*student*) are defined as $\mathbf{I}_{\mathcal{D}_s} = \text{Render}(\mathbf{G}, \{\mathbf{C}^t\}_{t=1}^L)$ and are supervised to match RGB frames $\mathbf{I}_{\mathcal{D}_{rgb}}$ (*teacher*), forming the self-distillation loop.

Multi-trajectory supervision. To maximize viewpoint coverage for the input image, we sample $V = 6$ camera trajectories per input image, as shown in Fig. 3. For each trajectory v , we construct a spatiotemporal cache $\{\mathbf{P}^{t,v}\}_{t=1}^L$ along with $L = 121$ camera poses $\{\mathbf{C}^{t,v}\}_{t=1}^L$. Passing these caches through video diffusion model $\mathcal{V}$ yields latents $\mathbf{z}^v$, which the teacher decodes into RGB frames $\mathbf{I}_{\mathcal{D}_{rgb}}^v = \mathcal{D}_{rgb}(\mathbf{z}^v)$. The 3DGS decoder $\mathcal{D}_s$ learns to fuse these multiple $\mathbf{z}^v$ into coherent Gaussians $\mathbf{G}$ while filling disoccluded regions. Through this self-distillation design, the 3DGS decoder $\mathcal{D}_s$ is trained entirely from synthetic supervision provided by $\mathcal{V}$ —with no real-world captures whatsoever. Operating in latent space makes aggregation across trajectories efficient, while the explicit Gaussian output $\mathbf{G}$ enforces geometric consistency and directly enables downstream simulation and real-time rendering.

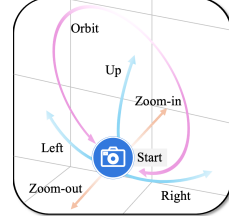


Figure 3: Six sampled camera trajectories that maximize view coverage.

4 FEED-FORWARD RECONSTRUCTION FROM MULTI-VIEW VIDEO LATENTS

Our feed-forward 3DGS decoder $\mathcal{D}_s$ transforms the synthesized multi-view latents from our video model $\mathcal{V}$ into an explicit 3D representation renderable from arbitrary viewpoints. We adopt 3D Gaussian Splatting (3DGS) (Kerbl et al., 2023) precisely because it combines an explicit representation with fast rendering. We detail our 3DGS decoder in Fig. 4 (right).

4.1 3DGS DECODER

Scaling Latent-Based 3D Reconstruction. Previous feed-forward reconstruction frameworks generate 3D Gaussians per pixel and are therefore tightly bounded by the number and resolution of input views they can handle. GS-LRM (Zhang et al., 2024e), for instance, operates on only 2–4 images at 512×512 resolution, while AnySplat (Jiang et al., 2025a) is trained with 24 images at

448 × 448 resolution. Our video model, by contrast, synthesizes $V \times L = 6 \times 121 = 726$ input views at 704 × 1280 resolution—far beyond the capacity of existing methods and well past the GPU memory limit. The root bottleneck is the attention mechanism over visual tokens, whose memory and compute scale with the number of pixels. We overcome it decisively: rather than scaling in pixel space, we operate directly in the compressed video latent space produced by our camera-controlled video diffusion model. The multi-view video latents are denoted as $\mathbf{Z} \in \mathbb{R}^{V \times L' \times C \times h \times w}$.

Architecture. The 3DGS decoder $\mathcal{D}_s$ first maps its inputs into the hidden dimension of the main reconstruction blocks and outputs per-pixel 3D Gaussian features $\mathbf{G} \in \mathbb{R}^{V \times L \times H \times W \times 14}$. Specifically, the inputs are the multi-view video latents $\mathbf{Z}$ and encoded Plücker embeddings $\mathbf{E}$, both with latent dimensions $\mathbf{Z}, \mathbf{E} \in \mathbb{R}^{V \times L' \times C \times h \times w}$. Each input component is first patchified to match the hidden dimension of the main reconstruction blocks. We do not require any additional visual encoder—only a spatial 2 × 2 patchification layer (Dosovitskiy et al., 2021) to transform the video latents into flattened tokens for the reconstruction network. The sum of both inputs is processed through the reconstruction blocks. We follow the design introduced in Long-LRM (Ziwen et al., 2024), i.e., one block consists of one Transformer (Vaswani et al., 2017) layer followed by seven Mamba-2 (Dao & Gu, 2024) layers. We repeat the block twice to get 16 layers with 512 hidden dimensions. Finally, a transposed 3D convolution maps the hidden representation to 14 Gaussian channels: 3D position (x, y, z) , scale (s_x, s_y, s_z) , rotation quaternion (q_w, q_x, q_y, q_z) , opacity α , and RGB (r, g, b) . Formally, the static decoder is expressed as $\mathbf{G} = \mathcal{D}_s(\mathbf{Z}, \mathbf{E})$.

Plücker embeddings. Raw Plücker embeddings are first computed from camera poses $\{\mathbf{C}^{t,v}\}$ and intrinsics as $\mathbf{E}^{\text{raw}} \in \mathbb{R}^{V \times L \times 6 \times H \times W}$. We reuse the RGB encoder $\mathcal{E}$ from the pre-trained VAE to encode Plücker embeddings. Specifically, we separately encode the 3-dimensional ray directions and the 3-dimensional cross product of ray directions and origins using $\mathcal{E}$, and concatenate latent along the channel dimension to obtain $\mathbf{E}^{\text{enc}} \in \mathbb{R}^{V \times L' \times 2C \times h \times w}$. A lightweight MLP maps $\mathbf{E}^{\text{enc}}$ to $\mathbf{E} \in \mathbb{R}^{V \times L' \times C \times h \times w}$, reducing the channel dimension by a factor of 2 to match video latent $\mathbf{Z}$.

4.2 LOSS FUNCTION

Image-based supervision. We supervise our 3DGS decoder with an image-based reconstruction loss. The reconstruction loss is split into a Mean Squared Error (MSE) loss $\mathcal{L}_{mse}$ and an LPIPS loss $\mathcal{L}_{lpips}$ (Zhang et al., 2018) using VGG (Simonyan & Zisserman, 2014) as a feature extractor.

Depth supervision. A decoder trained with RGB loss alone reliably collapses to flattened geometry. We eliminate this failure by additionally supervising the rendered depth maps with consistent video depth from the off-the-shelf ViPE system (Huang et al., 2025), using the scale-invariant depth loss $\mathcal{L}_{depth}$ of Long-LRM (Ziwen et al., 2024).

Opacity-based pruning. Similar to Long-LRM (Ziwen et al., 2024), we use an L1 regularization on the opacity $\mathcal{L}_{opacity}$ and remove the Gaussians with the lowest 80% opacity.

Total loss. Our total loss is computed as the weighted sum of all losses with weight factors λ_i

$$\mathcal{L} = \lambda_{mse} \mathcal{L}_{mse} + \lambda_{lpips} \mathcal{L}_{lpips} + \lambda_{depth} \mathcal{L}_{depth} + \lambda_{opacity} \mathcal{L}_{opacity} \quad (1)$$

where we set $\lambda_{mse} = 1.0$, $\lambda_{lpips} = 0.5$, $\lambda_{depth} = 0.05$, and $\lambda_{opacity} = 0.1$.

5 EXTENSION TO DYNAMIC 3D SCENES

Our approach extends to dynamic 3D scenes with only minimal changes, which we outline below.

Self-distillation for dynamic scenes. Our dynamic 3D setup closely follows the design of the static 3D counterpart. Instead of a single image, the video model takes a single video of length L with corresponding camera poses as input and generates multi-view video latents capturing the same underlying motion. For video inputs, we follow a protocol similar to the static case: sampling diverse text prompts with large language models, then generating videos using video diffusion models (Cosmos, 2025; Wan, 2025). The generated videos are then annotated with camera poses and depth maps using ViPE (Huang et al., 2025).

Dynamic 3DGS decoder architecture. We follow the bullet-time design of Liang et al. (2025b) and design our dynamic 3DGS decoder $\mathcal{D}_s$ to generate time-dependent 3D Gaussians for all the video

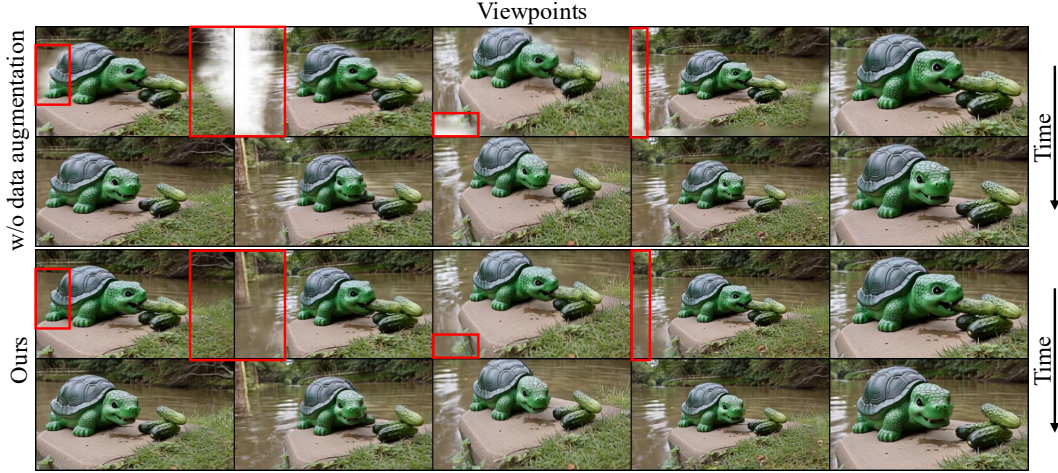


Figure 5: **Dynamic data augmentation.** Naively supervising the time-aware 3DGS decoder produces clear artifacts in the generated 3D Gaussians: early timesteps viewed from extreme poses show low opacity in regions the supervision signal never covers. Our motion-reversed augmentation ensures every timestep is observed with full spatial coverage, eliminating these low-opacity artifacts in the early timesteps.

frames. Specifically, the dynamic 3DGS decoder $\mathcal{D}_s$ closely follows the architecture of $\mathcal{D}_s$, except that we augment the $\mathcal{D}_s$ input with the encoded source and target time embeddings $\mathbf{T}^{\text{src}}, \mathbf{T}^{\text{tgt}} \in \mathbb{R}^{V \times L' \times C \times h \times w}$. The unencoded (raw) source times are assigned to each input frame, forming $\mathbf{T}^{\text{src, raw}} \in \mathbb{R}^{V \times L \times 1 \times H \times W}$, and the target time is $\mathbf{T}^{\text{tgt, raw}} \in \mathbb{R}^{V \times 1 \times 1 \times H \times W}$. Both are spatially repeated and augmented with a 2-dimensional sinusoidal embedding to expand the channel dimension to 3, then encoded with the RGB encoder $\mathcal{E}$ to latent dimensions $\mathbf{T}^{\text{src}}, \mathbf{T}^{\text{tgt}} \in \mathbb{R}^{V \times L' \times C \times h \times w}$. The target time is repeated along the latent temporal dimension to match L' . During training, a target time step is randomly sampled, and the corresponding 3D Gaussians are decoded and supervised across all trajectories generated with dynamic data augmentation. We fine-tune $\mathcal{D}_d$ from a pre-trained $\mathcal{D}_s$ with the patchification layers of $\mathbf{T}^{\text{src}}$ and $\mathbf{T}^{\text{tgt}}$ initialized to zeros. $\mathbf{T}^{\text{src}}$ and $\mathbf{T}^{\text{tgt}}$ are directly added to the sum of $\mathbf{Z}$ and $\mathbf{E}$. Formally, the dynamic decoder is defined as $\mathbf{G} = \mathcal{D}_d(\mathbf{Z}, \mathbf{E}, \mathbf{T}^{\text{src}}, \mathbf{T}^{\text{tgt}})$.

Dynamic data augmentation. In static 3D scenes, where time is not a factor, frames from any timestep can supervise the 3DGS renderings. For dynamic 3D scenes, however, the 3DGS changes over time, so only frames from the matching timestamp are valid for supervision. Left unaddressed, this restriction invites a trivial solution in which the 3DGS at a given timestep simply ignores information from frames at other timesteps (see Fig. 5).

We resolve this with a dynamic data augmentation strategy that balances supervision across timesteps. Early frames naturally correspond to viewpoints close to the input image, while later frames correspond to more distant viewpoints. To counter this imbalance, we augment the training data with paired supervision views for each timestep—one near and one far. Concretely, we reverse the input video in frame order and feed it into the video model $\mathcal{V}$, producing six additional multi-view sequences. Reversing these sequences back to the original motion order yields six trajectories that progress inward (from far viewpoints toward the input). Combined with the original six outward trajectories, this gives 12 supervision views per timestep. This augmentation is applied only during training: the reversed trajectories serve purely as extra supervision to prevent collapse from pruning artifacts and are not needed at inference time.

6 EXPERIMENTS

6.1 EXPERIMENTAL SETUP

Datasets. We train our model without any existing multi-view dataset; instead, we construct our own, the *Lyra* dataset, relying solely on our video model to supervise the 3DGS decoder. The 3D reconstruction setup uses 59,031 images and the 4D setup 7,378 videos. All data derive from diverse text prompts spanning indoor and outdoor environments, humans, animals, and both realistic

Table 1: **State-of-the-art comparisons.** Against prior single-image-to-3D generation methods on RealEstate10K, DL3DV, and Tanks-and-Temples, our method wins on every metric and every dataset. Best results in **bold**.

Method	RealEstate10K			DL3DV			Tanks-and-Temples		
	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$
ZeroNVS	13.01	0.378	0.448	13.35	0.339	0.465	12.94	0.325	0.470
ViewCrafter	16.84	0.514	0.341	15.53	0.525	0.352	14.93	0.483	0.384
Wonderland	17.15	0.550	0.292	16.64	0.574	0.325	15.90	0.510	0.344
Bolt3D	21.54	0.747	0.234	-	-	-	-	-	-
Ours	21.79	0.752	0.219	20.09	0.583	0.313	19.24	0.570	0.336



Figure 6: **Image-to-3DGS Generation.** Five views from our generated 3DGS scenes, demonstrating consistent, high-fidelity synthesis across viewpoints.

and imaginative content. We synthesize six camera trajectories for each image (3D) or video (4D), yielding 354,186 videos for 3D and 44,268 videos for 4D.

Baselines. We compare against the strongest state-of-the-art approaches for generative single-image-to-3D reconstruction, i.e., ZeroNVS (Sargent et al., 2024), ViewCrafter (Yu et al., 2024b), Wonderland (Liang et al., 2025a), and Bolt3D (Szymanowicz et al., 2025b). Since *no source code is available* for these methods to evaluate them on our out-of-distribution set, we compare directly against the quantitative results reported in their papers.

Evaluation. Following prior work, we evaluate our model on single-image-to-3D using RealEstate10K (Zhou et al., 2018), DL3DV (Ling et al., 2024b), and Tanks and Temples (Knapitsch et al., 2017), adopting the exact evaluation protocol of Wonderland (Liang et al., 2025a) and Bolt3D (Szymanowicz et al., 2025b). We report the standard reconstruction metrics: PSNR, SSIM, and LPIPS.

6.2 MAIN RESULTS

Quantitative results. Tab. 1 reports quantitative results for single-image-to-3D. Our method outperforms all previous works on every benchmark and across every metric, without exception. This establishes that the remaining path to even higher quality runs through stronger video generative models, from which our reconstructions will directly benefit. Additional quantitative comparisons on the *Lyra* dataset are given in Appendix C.

Qualitative results. We visualize novel-view renderings from our generated 3DGS scenes in Fig. 1 and Fig. 6; video results will be released upon acceptance. Our method generates high-quality novel-view content for 3D/4D scenes in unseen regions while remaining faithful to the input image/video. Qualitative comparisons on the *Lyra* dataset are shown in Appendix C.



Figure 7: **Ablations.** We render the same extreme novel viewpoint after image-to-3DGS generation, making the contribution of each component immediately visible; depth visualizations and further ablations are provided in Appendix C.4.

6.3 ABLATIONS

In Tab. 2, we validate each of our design choices by ablating the key components on diverse out-of-distribution prompts, comparing 3DGS renderings against the videos generated by our camera-controlled video diffusion model.

Real data only. Replacing self-distillation with training on only real multi-view datasets, i.e., RealEstate10K (Zhou et al., 2018) and DL3DV (Ling et al., 2024b), collapses generalization to out-of-distribution scenes—despite these being the datasets standard in prior work.

Self-distillation + real data. Strikingly, jointly training with self-distillation and real data yields no improvement over self-distillation alone. This is direct evidence that self-distillation is, by itself, diverse and consistent enough to learn a strong reconstruction model—real data adds nothing.

No depth loss. The depth loss eliminates flat geometry and even improves image-based metrics. We visualize depth maps in Appendix C.4.

No opacity pruning. We keep all Gaussians instead of pruning them by opacity. Learning to prune most of the 3D Gaussians makes the representation substantially more compact and improves visual quality. Rendering at $H = 704$ and $W = 1280$ takes 18ms with pruning versus 30ms without—a $1.67\times$ speedup.

No LPIPS loss. The LPIPS loss markedly improves robustness to input inconsistencies and sharpens the preservation of high-frequency details.

No multi-view fusion. A naive alternative generates 3D Gaussians for each camera trajectory independently and then fuses them into a single point cloud. Learning the multi-view fusion across trajectories with our reconstruction blocks, where each token attends to the others, is decisively better, as the ablation confirms.

No Mamba-2. Replacing our joint Transformer/Mamba-2 blocks with Transformer-only blocks lowers quality while being far more expensive. One forward pass with $V = 6$, $L = 121$, $H = 704$, and $W = 1280$ takes just 3213ms with our joint blocks versus 20922ms for Transformer-only blocks—a $6.5\times$ speedup.

No latent-based 3DGS. Pixel-space methods such as BTimer (Liang et al., 2025b) are capped at 12 input frames, whereas our latent-space design handles up to 726. Operating the 3DGS decoder in pixel space instead of latent space simply runs out of memory—underscoring that the latent formulation is essential, not merely convenient.

7 CONCLUSION

In this work, we introduced *Lyra*, a 3D and 4D generation framework that requires only a single image or video as input. Rather than collecting multi-view datasets, we cast camera-controlled video diffusion models as teachers within a self-distillation framework for a student 3DGS decoder. Our

Table 2: **Ablation study on the *Lyra* dataset.** Removing any component degrades quality, confirming that each design choice is necessary.

Method	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$
Ours	24.77	0.837	0.224
Data			
real data only	19.08	0.659	0.413
self-distill. + real data	24.74	0.823	0.236
Loss			
w/o depth loss	24.31	0.811	0.247
w/o opacity pruning	24.55	0.820	0.237
w/o LPIPS loss	23.74	0.766	0.370
Architecture			
w/o multi-view fusion	17.73	0.632	0.446
w/o Mamba-2	24.58	0.818	0.241
w/o latent 3DGS		OOM	

3DGS decoder operates in the latent space of the video model and directly reconstructs 3D Gaussians with no post-processing or optimization whatsoever. This design delivers strong generalization and coverage across diverse scenes, and our dynamic extension establishes 4D generation from monocular video. The scale and consistency of our generated scenes are bounded only by the capacity of the underlying camera-controlled video diffusion model; as these models improve, *Lyra* will directly inherit their gains, unlocking large-scale, consistent scene synthesis. Adapting auto-regressive techniques (Chen et al., 2024a) into our framework is a promising route to even larger-scale generation, and modelling motion and tracking information within the reconstruction network, as in concurrent work (Lin et al., 2025), offers a clear path to further improving visual motion quality.

8 ETHICS STATEMENT

Like all generative AI technologies, our method carries risks of misuse, including the production of misleading or synthetic 3D/4D content. Such risks acknowledged, the central goal of this research is to advance simulation, robotics, and embodied AI by providing scalable and controllable tools for data generation. We expect our most significant contributions to come through the following capabilities and insights.

- **Realistic simulation for embodied agents:** We deliver controllable, physically consistent camera motion within generated 3D and 4D environments, directly supporting the training and evaluation of agents that must perceive, navigate, and interact with complex scenes.
- **Scalable data engines:** By removing the need for multi-view capture and expensive per-scene optimization, our method produces high-fidelity, customizable scenes at scale—an ideal fit for research in robotics, reinforcement learning, and closed-loop simulation.
- **Deeper technical understanding:** As an academic study, this work provides a foundational case study of how camera control interacts with spatiotemporal generative models, informing research across computer vision, graphics, and embodied AI.

While we caution against deceptive or wrongful use of this technology—where misleading or unverifiable media could be created with realistic generative outputs—we are confident that established safeguards (e.g., provenance tracking, dataset documentation, and rigorous evaluation practices) will keep generative models advancing science and society in positive ways.

9 REPRODUCIBILITY STATEMENT

To fully reproduce our results and accelerate future research in this area, our training and inference code, model weights, and data for both 3D and 4D generation will be released upon acceptance.

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A ADDITIONAL DETAILS OF VIDEO DIFFUSION MODEL

This section provides additional detail on the GEN3C (Ren et al., 2025) video diffusion model backbone introduced in Sec. 3.1 and shows how we further strengthen its 3D consistency.

Conservative mask refinement. A key limitation of GEN3C’s forward warping is background leakage: occluded object parts in the source view (e.g., the lion body in Fig. 8) become visible from novel viewpoints yet are not flagged in the disocclusion mask. This causes the video diffusion model to complete the foreground incompletely, as shown in Fig. 8 (second and fourth columns).

We fix this with a conservative strategy: since the geometry of occluded regions in $\mathbf{P}^{t,v}$ is unknown, we treat all such areas as occupied and derive refined disocclusion masks accordingly. This lets the video model reason about potentially visible foreground regions that standard point-based rendering misses.

Our approach constructs a triangular mesh from camera-space points $\mathbf{P}_c^{t,v} = \mathbf{C}^t \mathbf{P}^{t,v}$ by connecting spatially adjacent pixels, similar to Hu et al. (2021):

$$\mathcal{M} = \bigcup_{(u,v)} \{ \triangle(\mathbf{p}_{u,v}, \mathbf{p}_{u+1,v}, \mathbf{p}_{u,v+1}), \triangle(\mathbf{p}_{u+1,v}, \mathbf{p}_{u+1,v+1}, \mathbf{p}_{u,v+1}) \} \quad (2)$$

where $\mathbf{p}_{u,v}$ denotes the 3D point at pixel (u, v) in $\mathbf{P}_c^{t,v}$. This surface mesh serves as a boundary between visible and invisible regions. For each rendered pixel, we compare the standard point-based depth $\mathbf{D}^{t,v}$ with the mesh-interpolated depth $\mathbf{D}_{\mathcal{M}}^{t,v}$ obtained via ray-surface intersection:

$$\mathbf{M}^{t,v}(u, v) = \begin{cases} 0 & \text{if } \mathbf{D}_{\mathcal{M}}^{t,v}(u, v) < \mathbf{D}^{t,v}(u, v) - \epsilon \\ \mathbf{M}_{\text{orig}}^{t,v}(u, v) & \text{otherwise} \end{cases}, \quad (3)$$

where $\mathbf{M}_{\text{orig}}^{t,v}$ denotes the original disocclusion mask from forward warping and ϵ is a small tolerance factor. When the mesh surface is closer than the original points, we conservatively mask out these regions as potential foreground disocclusions. This yields markedly more reliable guidance for video generation, especially under large viewpoint changes, as shown in Fig. 8 (third and fifth columns).

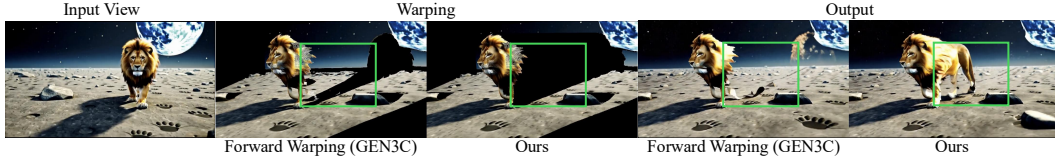


Figure 8: **Rendering function comparison.** Our model builds upon GEN3C (Ren et al., 2025), which uses forward-warped images as camera-control conditioning. We contrast forward warping (left) with our conservative rendering function (right). The highlighted region is incorrectly filled with background pixels under forward warping but is correctly masked by ours, enabling proper completion.

B MODEL DETAILS

Progressive training. Directly training the 3DGS decoder on 6 trajectories of 121 frames at 704×1280 is prohibitively expensive, so we train our network progressively, as detailed in Tab. 3, and initialize the dynamic model from the final static pre-trained model. Total training takes just 6 days on 8 NVIDIA A100 80 GB GPUs. We use gsplat (Ye et al., 2025) as the 3DGS implementation.

Subsampling Gaussians. Generating pixel-aligned 3D Gaussians scales linearly with spatial and temporal resolution. Our generated scenes span 726 frames at 704×1280 resolution, which would require 654,213,120 per-pixel Gaussians. We therefore generate a single 3D Gaussian per 8×8 spatial neighborhood, cutting the total by a factor of 64 to 10,222,080, and our opacity-based pruning further reduces it to 2,044,416—yielding a far more compact output.

Joint Transformer-Mamba blocks. We adopt the reconstruction block design of Long-LRM (Ziwen et al., 2024): one block is a single Transformer layer followed by seven Mamba-2 layers, repeated

Table 3: **Progressive training setup.**

Stage	$H \times W$	L	V	S	B	Steps
Static						
1	176×320	17	1	17	4	10k
2	176×320	49	1	49	4	2.5k
3	352×640	49	1	49	2	2.5k
4	704×1280	49	1	49	1	2.5k
5	704×1280	121	1	9	1	57.5k
6	704×1280	121	1-6	9	1	7k
Dynamic						
7	704×1280	121	6	12	1	10k

twice for 16 layers with 512 hidden dimensions. Notably, our network is far smaller than those of Long-LRM (Ziwen et al., 2024) and Wonderland (Liang et al., 2025a), which use 24 layers with 1024 hidden dimensions—yet it delivers state-of-the-art results. While pure Transformers are typically more powerful, they are significantly slower in both training and inference than Mamba-2. Because we trained both configurations under the same budget, we attribute the gap to optimization: Transformer-only blocks would need considerably more training iterations to catch up in rendering quality.

Camera encoding. We use Plücker embeddings for camera encoding, following prior works (Kant et al., 2024; Zhang et al., 2024e). The embeddings are computed at full spatial and temporal resolution and then projected into the video latent space using the pre-trained RGB encoder. Since the RGB encoder expects 3 input channels but Plücker embeddings are 6-dimensional, we separately encode the 3-dimensional ray directions and the 3-dimensional cross product of ray directions and origins. The resulting latent encodings are then concatenated along the channel dimension.

Time encoding. Similarly, to make the 1-dimensional time input compatible with the 3-channel input expected by the RGB encoder, we concatenate a 2-dimensional sinusoidal embedding to the original time value along the channel dimension. We then replicate these values across the spatial dimensions before encoding the time into the video latent space. This approach allows us to independently encode both the source and target times into the compressed latent representation.

Dynamic data augmentation. We visualize our dynamic data augmentation procedure for two example camera trajectories in Fig. 9. The procedure creates pairs of supervision views that cover the scene from the same motion state but different extreme viewpoints.

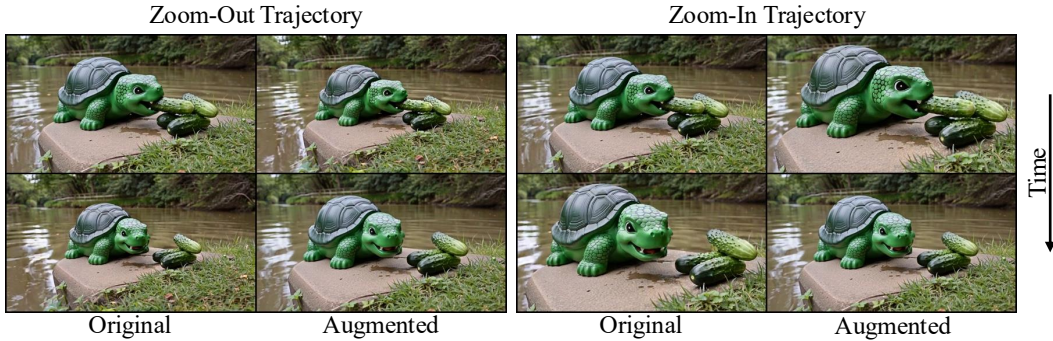


Figure 9: **Dynamic data augmentation videos.** Augmenting the supervision data with a motion-reversed video ensures every timestep is observed with full spatial coverage, eliminating low-opacity artifacts in the early timesteps. We show two example trajectories, zoom-out and zoom-in, together with their augmented videos, whose camera motion is flipped.



Figure 10: **Image-to-3DGS Comparison.** We compare against BTimer (GEN3C) across five views from generated scenes. Our method produces markedly fewer artifacts and clearly higher fidelity.

C ADDITIONAL EVALUATION

C.1 ADDITIONAL BASELINE DETAILS

We additionally benchmark our model against BTimer (Liang et al., 2025b), a recent approach for joint 3D and 4D reconstruction from posed images or videos. We asked the BTimer authors to run experiments for us on the same scenes as *Lyra*. Since BTimer is purely regression-based, we pair it with our GEN3C (Ren et al., 2025) video diffusion backbone for a fair, like-for-like comparison. Specifically, we use the same camera trajectories as in our model and provide RGB-decoded videos from GEN3C as input to BTimer. BTimer operates in pixel space and requires 12 input images. However, as discussed in the main paper, pixel-space models such as BTimer run out of memory when directly using the 726 high-resolution frames generated by GEN3C. To address this, we uniformly subsample GEN3C frames. For static evaluations, we select 12 frames that evenly cover the available viewpoints. For dynamic evaluations, we sample 11 frames to span the viewpoint range and additionally include the bullet-time frame corresponding to the motion being reconstructed. This ensures that the target motion state is always present as a reference. We denote this integrated baseline as BTimer (GEN3C), i.e., BTimer combined with GEN3C.

C.2 ADDITIONAL STATIC 3D EVALUATION

For image-to-3DGS generation, we show qualitative comparisons with BTimer (GEN3C) in Fig. 10 and quantitative results in Tab. 4a. Our method clearly delivers higher quality with far fewer artifacts.

C.3 DYNAMIC 3D EVALUATION

BTimer (GEN3C) comparison. For dynamic 3D evaluation, we compare against BTimer (GEN3C), as discussed in Sec. C.1, on 100 out-of-distribution videos from our dynamic *Lyra* dataset. We crop the videos to 512×512 for a fair comparison, since BTimer was primarily trained at that resolution. As Tab. 4b shows, our method decisively outperforms BTimer (GEN3C).

C.4 DEPTH VISUALIZATION

We visualize depths from our generated 3DGS in Fig. 11. Depth supervision reliably prevents flat geometries while preserving visual quality.

Table 4: **Quantitative results on the static and dynamic *Lyra* datasets.** Our method outperforms BTimer (GEN3C) by a wide margin in both settings; best results in **bold**.

Method	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$
BTimer (GEN3C)	16.32	0.580	0.427
Ours	24.92	0.834	0.183

(a) Comparison on static *Lyra* dataset.

Method	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$
BTimer (GEN3C)	20.29	0.687	0.315
Ours	23.07	0.779	0.231

(b) Comparison on dynamic *Lyra* dataset.

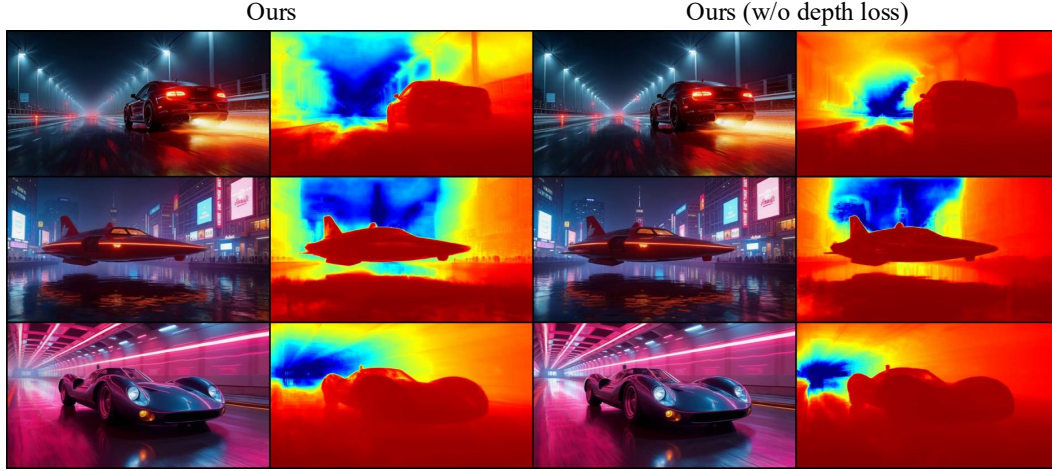


Figure 11: **Depth loss ablation.** 3DGS renderings and their corresponding depth maps. Adding depth supervision reliably prevents flat geometries while fully preserving visual quality.

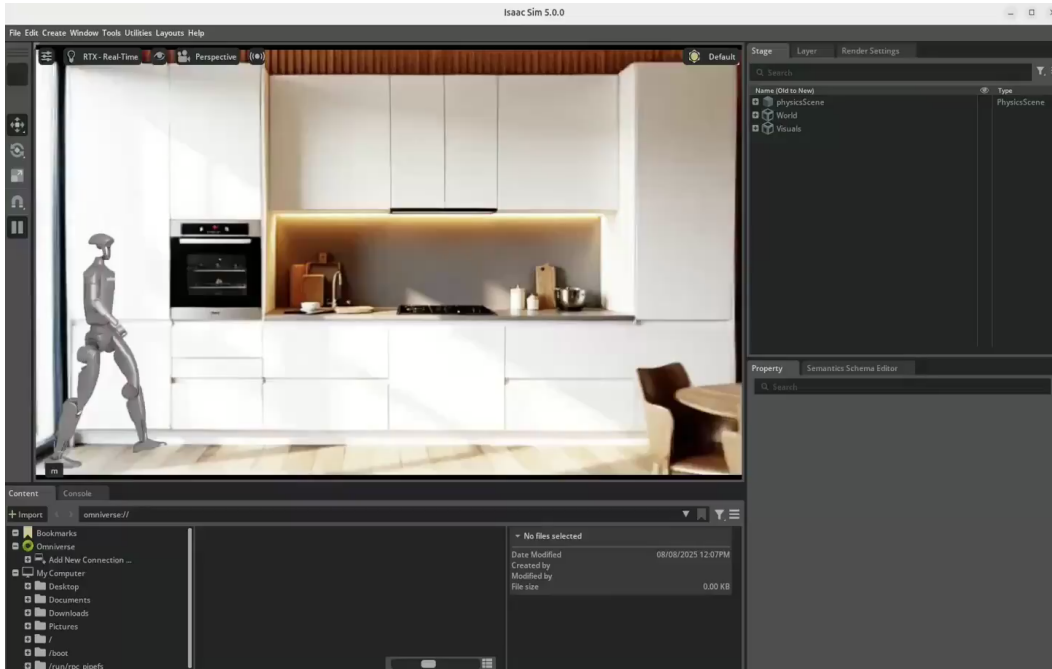


Figure 12: **Robot Simulation.** A frame of a robot simulation running in the Isaac Sim 5.0 framework directly on a 3DGS scene generated by our method.

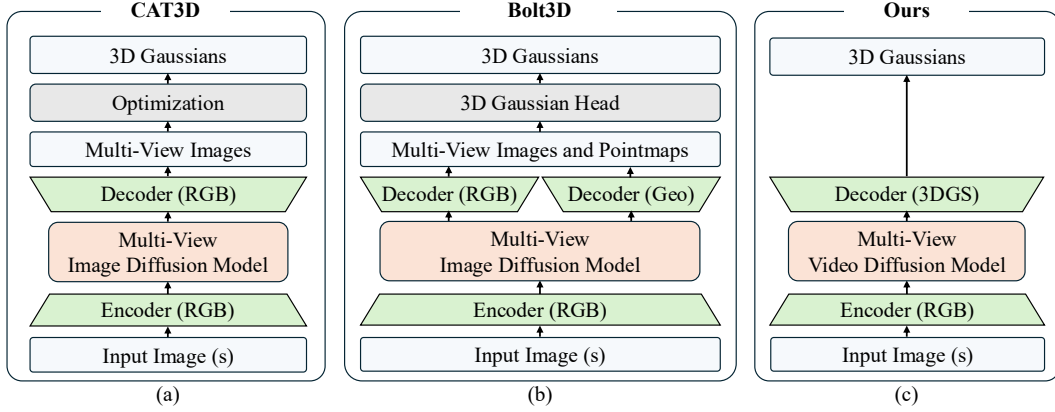


Figure 13: **Approaches for 3D Generation.** CAT3D (Gao et al., 2024b) proposes a multi-view image diffusion model that outputs images from novel viewpoints, which are then reconstructed into 3D Gaussians via optimization. Bolt3D (Szymanowicz et al., 2025b) fine-tunes CAT3D to output pointmaps with a geometry autoencoder and, instead of optimization, predicts the 3D scene with a feed-forward 3D Gaussian head. Our work goes further: it builds on a pre-trained camera-controlled video diffusion model and *directly* decodes the multi-view video latents into 3D Gaussians.

D ROBOT SIMULATION

A primary motivation for building a diverse 3D generator is to synthesize simulation environments for autonomous agents, and we demonstrate this end to end. We first generate 3D scenes from text with our model, export the 3D Gaussians as a .ply file, and convert them to .usdz format using 3DGUT (Wu et al., 2025a) and its export function. The resulting .usdz file imports directly into the NVIDIA Isaac robot development platform as a physically based virtual environment, where AI-based robots can be trained and tested under diverse conditions. Early demo results will be released upon acceptance; we include a screenshot of the interface in Fig. 12.

E RELATED WORK

In this section, we discuss additional related work and highlight how our approach differs from prior work, namely CAT3D (Gao et al., 2024b) and Bolt3D (Szymanowicz et al., 2025b), in Fig. 13.

3D generation. Early approaches to 3D generation primarily focused on single object categories, extending GANs into the 3D domain by leveraging neural renderers as an inductive bias (DeVries et al., 2021; Chan et al., 2022; Or-El et al., 2022; Schwarz et al., 2022; Bahmani et al., 2023b). With the introduction of CLIP-based supervision (Radford et al., 2021), the field advanced toward more flexible and diverse asset creation, enabling both text-driven generation and editing (Chen et al., 2018; Jain et al., 2022; Sanghi et al., 2022; Jetchev, 2021; Gao et al., 2023; Wang et al., 2022). More recently, diffusion models have significantly improved quality by replacing CLIP guidance with Score Distillation Sampling (SDS) (Poole et al., 2023; Wang et al., 2023b; Lin et al., 2023a; Chen et al., 2023b; Liang et al., 2023; Wang et al., 2023a; Li et al., 2024e; He et al., 2024b; Ye et al., 2024; Liu et al., 2024b; Yu et al., 2023; Katzir et al., 2024; Lee et al., 2024a; Sun et al., 2024a).

To further enhance structural coherence, several methods enforce multi-view consistency by generating scenes from different perspectives (Lin et al., 2023b; Liu et al., 2023; Shi et al., 2024; Feng et al., 2024a; Liu et al., 2024a; Kim et al., 2023; Voleti et al., 2024; Höllein et al., 2024; Tang et al., 2024c; Gao et al., 2024b; Wang et al., 2025f; Kant et al., 2025; Yuan et al., 2025; Ren et al., 2024b), while others adopt iterative inpainting as a strategy for scene synthesis (Höllein et al., 2023; Shriram et al., 2024). Another line of work explores lifting 2D images into 3D using NeRF (Mildenhall et al., 2020), 3D Gaussian Splatting (Kerbl et al., 2023), or mesh-based representations coupled with diffusion models (Chan et al., 2023; Tang et al., 2023a; Gu et al., 2023; Liu et al., 2024c; Yoo et al., 2023; Tewari et al., 2023; Qian et al., 2024b; Long et al., 2024; Wan et al., 2024; Szymanowicz et al., 2023; Lu et al., 2024).

Feed-forward 3D models. Complementary efforts investigate fast, feed-forward techniques that directly predict 3D content from images or text inputs (Hong et al., 2024; Li et al., 2024b; Xu et al., 2024e;d; Zhang et al., 2024a; Han et al., 2024; Jiang & Wang, 2024; Xie et al., 2024a; Tang et al., 2024b; Tochilkin et al., 2024; Qian et al., 2024a; Szymanowicz et al., 2024; 2025a; Liang et al., 2025a; Szymanowicz et al., 2025b; Schwarz et al., 2025; Yang et al., 2025; Zhang et al., 2025a). Unlike our method, however, all of these are confined to static 3D scenes. Other methods target specific domains such as faces (Kirschstein et al., 2025). Concurrent work (Liang et al., 2025b; Xu et al., 2025) handles real scenes but cannot cope with diverse generated scenes or large viewpoint changes—precisely the regime our method masters.

4D generation. Recent years have seen rapid progress in 4D generation, i.e., dynamic 3D scene synthesis. Many approaches leverage input text prompts or images as guidance. Following the introduction of large-scale generative models for this task (Singer et al., 2023), subsequent works have made notable advances in both visual fidelity and motion quality (Ren et al., 2023; Ling et al., 2024a; Bahmani et al., 2024b; Zheng et al., 2024b; Gao et al., 2024a; Yang et al., 2024a; Jiang et al., 2024b; Miao et al., 2024; Li et al., 2024a; Zhang et al., 2024g; Yuan et al., 2024; Jiang et al., 2024a).

While text conditioning is widely used, alternative methods aim to lift 2D images or videos into dynamic 3D scenes (Ren et al., 2023; Zhao et al., 2023; Yin et al., 2023; Pan et al., 2024; Zheng et al., 2024b; Ling et al., 2024a; Gao et al., 2024a; Zeng et al., 2024b; Chu et al., 2024; Wu et al., 2024; Yang et al., 2024b; Wang et al., 2024c; Feng et al., 2024b; Sun et al., 2024c; Zhang et al., 2024c; Ren et al., 2024a; Lee et al., 2024b; Li et al., 2024c; Van Hoorick et al., 2024; Uzolas et al., 2024; Chai et al., 2024; Liang et al., 2024; Li et al., 2024d; Xie et al., 2024b; Li et al., 2025a; Wang et al., 2025e; Zhu et al., 2025; Zhou et al., 2025; Hu et al., 2025; Park et al., 2025). Other efforts explore incorporating physics priors into generation pipelines (Lin et al., 2024; Huang et al., 2024b; Zhang et al., 2024f; Kiray et al., 2025), or enhancing motion controllability with template-driven approaches (Zhang et al., 2024d; Sun et al., 2024b; Chen et al., 2024b). A parallel direction emphasizes compositional and interactive 4D generation (Bahmani et al., 2024a; Xu et al., 2024b; Cao et al., 2024; Yu et al., 2024a; Zeng et al., 2024a; Zhu et al., 2024).

Another strand of research extends 3D GAN frameworks into the 4D setting by training directly on 2D video data (Bahmani et al., 2023a; Xu et al., 2023). Such methods are constrained by small-scale, single-category datasets that limit generalization, and most remain object-centric, neglecting background elements. Consequently, their visual fidelity falls well short of the photorealism of recent video generation models—the very models our method harnesses.

Camera-conditioned video models. Recent progress has focused on incorporating camera control into video diffusion models. The pioneering work MotionCtrl (Wang et al., 2024e) introduced camera conditioning by augmenting pre-trained video models (Chen et al., 2023a; Blattmann et al., 2023) with extrinsic matrices. Subsequent efforts (He et al., 2024a; Xu et al., 2024c; Kuang et al., 2024; Ju et al., 2025; He et al., 2025; Li et al., 2025c; Wang et al., 2025c; 2024d; Lei et al., 2025; Liu et al., 2025) enhanced conditioning by representing cameras using Plücker coordinates, while others use follow MotionCtrl to simply use poses flattened (Wang et al., 2025b). Another line of research (Hu et al., 2024; Xiao et al., 2024; Ling et al., 2024c; Hou et al., 2024) enabled camera motion control without introducing additional trainable parameters. Despite these advances, all of the above rely on U-Net-based architectures. Other approaches use low-rank approaches for camera control (Zhang et al., 2025b).

Several recent works continue to extend U-Net-based approaches for camera control (Zhao et al., 2024; Xu et al., 2024a; Zheng et al., 2024a; Zhang et al., 2024b; Yu et al., 2024b; Lei et al., 2024), while others begin exploring diffusion transformers. Cheong et al. (Cheong et al., 2024) propose one such transformer-based framework, though its scene and visual quality remain limited. DimensionX (Sun et al., 2024d) introduces joint space-time control in diffusion transformers, but relies on pre-defined, non-continuous camera trajectories. Other works (Huang et al., 2024a; Mai et al., 2025; Jiang et al., 2025b) instead explore pose and geometry estimation with a video DiT. CAT4D (Wu et al., 2025b) proposes a multi-view video diffusion model derived from fine-tuning a multi-view backbone. Large-scale synthetic data generation further boosts performance, as demonstrated by SynCamMaster (Bai et al., 2025b) and ReCamMaster (Bai et al., 2025a), which achieve strong results on camera-controlled video generation using transformer-based architectures. Finally, 4DiM (Watson et al., 2024) trains a space-time diffusion model from scratch for novel view synthesis from a single input image. Another line of work edits camera movements of videos with underlying motion using camera retargeting (Seo

1512 et al., 2025; Ma et al., 2025). One popular direction has been to use a spatio-temporal cache, e.g.,
1513 using point clouds or optical flow, to condition the camera-controlled video generation (Wu et al.,
1514 2025c; Ren et al., 2025; Wang et al., 2025d; Cao et al., 2025; YU et al., 2025; Jin et al., 2025; Xing
1515 et al., 2025; Yan et al., 2025; Li et al., 2025b; Gu et al., 2025).
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